

Virtualisation for All

The ability to run multiple copies of different operating systems on one physical machine is already quite predominant in the IT world, and has been for several decades. However, systems with microprocessors that provide hardware acceleration have greatly spurred virtualisation and made it a real pleasure.



Imagine a large rack holding dozens of workstations and servers. And compare that with a small 2U server that has all that infrastructure running on it. Progress? Yes, definitely! You save on physical space, consume much less electricity, and third, increase management efficiency by having all these systems in one place. So let us transform this dream into a reality with the open source virtualisation software Proxmox.

Prerequisites

Originally, I had several projects deployed on VMware Server 2.0—installations of Linux and Windows systems. Everything was convenient for me, because I had already resolved the first two factors—space and electricity bills—as all the systems were on virtual infrastructure. However, after I added more Linux projects to the host, VMware became non-operational—I couldn't log in to the VMware Web-based control panel to see what was happening, not even by means of a native client. Second, simultaneous disk-intensive operations inside two

virtual machines greatly affected other containers as well, resulting in slow responses from all machines. This was not a big problem, actually, but was irritating when I couldn't get immediate access to any virtual container.

A big motivation was the purchase, back then, of a new blade system with hardware virtualisation support at the CPU level, and the appearance of kernel-based virtualisation support in Linux.

Proxmox

In brief, Proxmox is a Debian-based distribution that comes with a specially compiled kernel, including support for Kernel-based Virtual Machine (KVM) and OpenVZ para-virtualisation from the Parallels company. You can thus run any x86-based OS with acceleration (thanks to technology like Intel VT-x or AMD-V) as a QEMU image, and also launch Linux distributions in isolated containers. The latter, for instance, results in a mere 1-3 per cent loss in computing performance compared to the host system.